Embedded Linux & System Software Interview Handbook

Phase 8 – Performance Analysis, Tracing & Kernel Debugging

This phase focuses on profiling Linux systems, tracing kernel execution and debugging production issues using industry-standard tools.

Chapter 15 – Performance Analysis, Tracing & Kernel Debugging

Topics Covered

• Linux Performance Fundamentals

• CPU Profiling

• Memory Profiling

• I/O Analysis

• Context Switch Analysis

• Interrupt Latency

• perf

• ftrace

• trace-cmd

• Kernel Tracepoints

• Dynamic Debug

• eBPF

• bcc Tools

• bpftrace

• SystemTap

• LTTng

• Flame Graphs

• strace

• ltrace

• gdb

• kgdb

• Kernel Panic Analysis

• Oops Analysis

• Crash Utility

• kdump

• Core Dump Analysis

• lockdep

• kmemleak

• KASAN
• KCSAN
• UBSAN
• Memory Leak Detection
• Race Condition Debugging
• Performance Tuning
• Production Case Studies
• Senior Interview Questions

Learning Outcomes

After completing this phase you will be able to trace kernel execution, diagnose performance bottlenecks, analyze crashes, debug memory and concurrency issues, and optimize Linux systems for production deployments.

Phase	Coverage
Phase 1	Core Linux Kernel Fundamentals
Phase 2	Memory & Process Management
Phase 3	Filesystems & Storage
Phase 4	Linux Networking
Phase 5	Interrupts, Timers & Deferred Work
Phase 6	Advanced Device Drivers
Phase 7	Linux Security & Secure Boot
Phase 8	Performance, Tracing & Kernel Debugging
Estimated Size	250–350 pages
Target	Senior Embedded Linux / Kernel Interviews